Roll Number:_____ Name.....

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science &Engineering

B. E.(COE/CSE) : MST	**Course Code: UCS510/UCS319**
	Course Name: Computer Architecture and Organization
26/Sept/2025	**3:00 AM – 5:00PM**
Time: 2 Hours, M. Marks: 28	**Faculty: SHI/AAS/YAS/JPK/ABD**

Note: ***All questions carry equal marks. Attempt all questions in order.***
Assume the missing data. Calculators are not allowed

Q1.(A)	The dean, the department chair, the teaching assistant, and the dorm social chair each use the auditorium from time to time. Unfortunately, they occasionally conflict, leading to disasters such as when the dean's fund-raising meeting with trustees happened at the same time as the dorm's black light party. You have been called to design a room reservation system. The system has four inputs A3, A2, A1, A0 and four outputs Y3, Y2, Y1, Y0. Each user asserts his input when he requests the auditorium for the next day. The system asserts at most one output, granting the auditorium to the highest-priority user. The dean (input A3) has highest priority, followed by the department chair (A2), then the teaching assistant (A1), and finally the dorm social chair (A0). *(use **SOP K-MAP** whenever required.)* (i) Write the truth table for the system. (ii) Write Boolean equations for output variables using SOP K-map. (iii) Sketch a circuit diagram for final function.	4	CO1	L3
Q1.(B)	Convert the following numbers into IEEE-754 double-precision (64-bit) floating point representation. (write final answer in hexadecimal) i. $(A4.75)_{16}$ ii. $(-0.09375)_{10}$	3	CO1	L1
Q2.(A)	A 16-bit register, R1, holds the signed decimal value -100. The system uses two's complement representation for negative numbers. Perform Arithmetic Left Shift on R1 by 2 bits. Give: a. The initial binary value in R1. b. The resulting binary value after the shift. c. The decimal equivalent of the final binary value.	3	CO1	L3
Q2.(B)	Design an ALU circuit with selection variable S_2, S_1, and S_0 and two 2-bit data inputs A and B. The circuit generates the following four arithmetic and shift micro-operations in conjunction with the input carry C_{in}. (***Note:*** *Draw complete circuit **[Do not use any black box diagrams]** for Arithmetic and Shift micro-operations given below*).	4	CO1	L6

S2	S1	S0	Cin	Micro-operation
0	0	0	0	A+B
0	1	1	1	A-B
1	0	0	X	Logic shift left
1	1	1	X	Logic shift right

<table>
<tr><td>Q3.(A)</td><td>A 16-bit basic computer's instruction format has three parts: indirect bit (I = 0), decoder operation (D 7 = 1 1 1), and register operations. The bit numbers rB 9 , rB 6 , rB 3 and rB 1 specify CMA, CIL, SNZ, and SZE register reference instructions, respectively. The instruction is stored at the memory location 849. The initial content of the accumulator (AC) and E flipflop is E48D and 0 (all content is in hexadecimal). Determine the content of PC, AR, IR, AC, E, and sequence counter (SC) after the execution of these instructions. Consider that each register-reference instruction is independently executed. Give the answer in a table as referenced below.

<table><tr><th>Instruction</th><th>PC</th><th>AR</th><th>IR</th><th>AC</th><th>E</th><th>SC</th></tr><tr><td>CMA</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>CIL</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>SNZ</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>SZE</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table></td><td>4</td><td>CO2</td><td>L3</td></tr>
<tr><td>Q3.(B)</td><td>An interrupt occurs while a computer system is executing a program, it causes the Interrupt Enable (IEN) flag = 1, the Input Flag (FGI) = 1, and the Output Flag (FGO) = 0.
a) Draw a flowchart and explain how the control unit will respond in a sequence of steps.
b) What happens to the Program Counter (PC) and the return address when the interrupt is acknowledged? Write your answer using suitable micro-operations.
c) After branching to location 1, what changes are made to the IEN and R registers?</td><td>3</td><td>CO2</td><td>L2</td></tr>
<tr><td>Q4.(A)</td><td>Assume 5-bit registers that hold the signed numbers. Show the step-by-step multiplication process using the Booth Algorithm where multiplicand is $(-9)_{10}$ and multiplier is $(-10)_{10}$.</td><td>3</td><td>CO3</td><td>L5</td></tr>
<tr><td>Q4(B)</td><td>Show the contents in hexadecimal of registers PC, AR, IR, AC, DR, E, and SC of the 16-bit basic computer when an instruction is fetched from memory and executed. The initial content of PC is A9F. The content of memory at address A9F is 9C2B. The content of memory at address C2B is 158D. The content of memory at address 58D is FA9F. The content of AC is the complement of DD80. Assume the initial bit of E as 0. Give the answer in a table for initial content and at each clock pulse with seven columns, one for each register and a row for each timing signal. (All content is in hexadecimal.)</td><td>4</td><td>CO2</td><td>L2</td></tr>
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